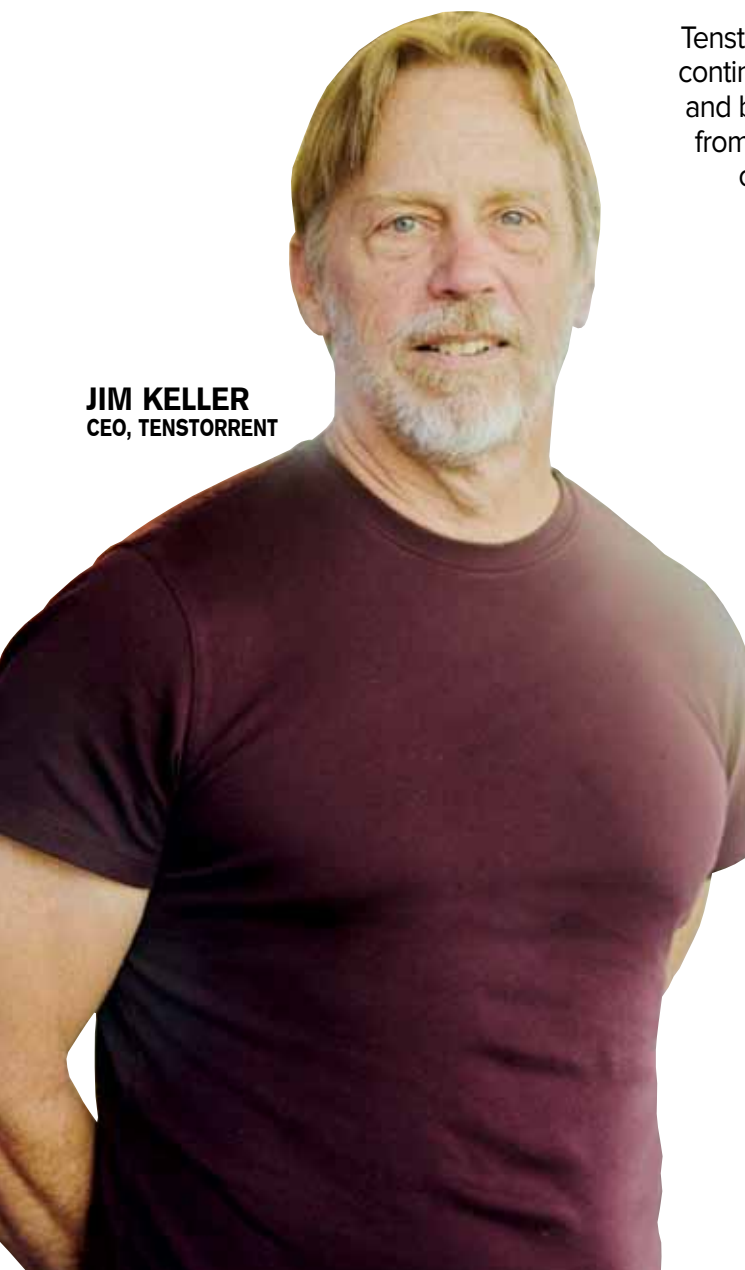


“AI PROCESSOR CHIPLETS Enable Small Companies TO SOLVE A HUGE PROBLEM By Breaking It Into Smaller Problems”



JIM KELLER
CEO, TENSTORRENT

Tenstorrent CEO Jim Keller believes that AI processors will continue to get bigger so that more people can participate and build high-performance AI models. Yashasvini Razdan from Electronics For You spoke to Keller to gain his views on the future of AI processors and RISC-V technology

How would you explain your new technology?

We are making a novel artificial intelligence (AI) computer. AI computers do massive amounts of dense mathematical computation and perform a large number of operations.

Computation is similar to how the brain works. Your brain is made out of around 10,000 groups of cells that talk to each other, locally, and then they talk to each other broadly. You can think of AI problems as very intense mathematics. Computations are done in the brain when many cells exchange information with each other, and they just keep doing it to make the brain work.

We're building a computer that can do a lot of mathematics, but also communicate the results of each step of the calculations. That kind of representation is trainable and leads to interesting results with AI software.

To sum it up, we are building hardware that has a large number of independent processors and we're building software that can take the company's AI program and spread it across those processors, and those processors can talk to each other, natively. This gives us an advantage on how to write the code and also advantages on the cost of the computer and the cost of the system.

What is the advantage of having multiple chips instead of a single processor?

A single server chip is big and very expensive. It's a single piece of silicon with multiple functions on that one piece. If we divide that chip into four pieces, each piece, called a chiplet, is smaller. You can now put the memory interface on one piece, the network interface on another piece, the IO